

Haoran (Alex) Yu

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EDUCATION

Boston University Metropolitan College

Master of Science in Applied Data Analytics

Boston, USA

Expected Entry: January 2026

University of Toronto, St. George Campus

Major in Mathematics (Minors in Statistics and Economics), HBS

Toronto, Canada

September 2021 – July 2025

- Relevant Courses:** Linear Algebra, Advanced Calculus, Discrete Mathematics, Group Theory, Real Analysis, Complex Analysis, Mathematical Modelling, Nonlinear Optimization, Probability and Statistics, Data Analysis Methods, Machine Learning, Statistical Survey Sampling, Financial Economics, Financial Mathematics, Micro and Macro Economics

Projects

Credit Risk Analysis of Enbridge Inc.

Independent Project

Toronto, Canada

March 2025 – April 2025

- Based on the Merton Model, used the 15-year Canadian government bond yield as the risk-free rate and annualized volatility of Enbridge's stock price (2010–2025) to iteratively solve for asset value volatility.
- Estimated the firm's distance to default and probability of default over the next 15 years.
- Built a two-step transition probability matrix assuming a 50% recovery rate and assessed credit risk based on the bond spread between Enbridge and U.S. Treasury bonds.
- Created a one-page A4 poster presenting the project's background, methodology, key findings, and conclusions.

Loan Default Risk Prediction Using Machine Learning

Independent Project

Toronto, Canada

August 2024 – December 2024

- Used the Lending Club dataset and performed data cleaning, missing value handling, and feature encoding using the pandas library.
- Explored variable distributions and correlations using Matplotlib and pandas to identify key predictors of loan defaults.
- Built and tuned a Random Forest model using scikit-learn with cross-validation to optimize performance.
- Evaluated model performance through confusion matrix, classification report, and ROC/AUC curve analysis.
- Entire pipeline developed and visualized in Jupyter Notebook.

Vehicle Insurance Premium Modeling

Team Leader and Group Project

Toronto, Canada

July 2024 – August 2024

- Led team management and project planning, and independently developed core modeling code.
- Used R language to construct linear regression models based on real-world Spanish auto insurance data.
- Applied Box-Cox, square root, logarithmic transformations, and ridge regression for model optimization.
- Conducted statistical tests including F-test, partial F-test, Cook's Distance, and R-squared evaluations to ensure model reliability.
- Authored a complete technical report in LaTeX detailing methodology, results, and visual analysis.

Internship Experience

Shaanxi Jintai Hengye Real Estate Co., Ltd.

Real Estate Investment Expansion Intern

Shaanxi, China

June 2023 – September 2023

- Assisted in collecting and systematically organizing land auction transaction data as well as legal and financial documents related to equity acquisitions
- Participated throughout the early stages of project development, including market research, SWOT competitive analysis, preliminary planning design, financial modeling, and risk assessment
- Contributed to the drafting of feasibility study reports under the guidance of senior analysts, gaining hands-on experience in core processes and professional methods of commercial project analysis

Programming & Tools:

Python and libraries (NumPy, pandas, Matplotlib, scikit-learn, SciPy), DEAP (Distributed Evolutionary Algorithms in Python), R and tidyverse, Git, LaTeX, SQL